

Addressing Mechanisms and Heterogeneity in Change with Usable Tools

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April 20, 2026



PennState

<https://github.com/jeksterslab/talks20260420PSUHDFSJobtalk>

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- Good morning, and thank you for the opportunity to be here.
- At the center of my research program is a simple goal: to understand **dynamic processes** in *development, family life, and health* in ways that are **substantively meaningful** and **methodologically usable**.
- In particular, I study *how change unfolds over time, how people differ in those processes, and how we can build tools that help researchers study those questions well*.
- What draws me to HDFS is that these questions are inherently **contextual** and **multilevel**, showing up in *individuals, relationships, families, and everyday settings*.
- Today, I will walk through three recent studies that reflect that agenda.
- Across them, the common thread is **methodology in service of substantive questions**.
- I will end with a few future directions I would be especially excited to pursue at Penn State HDFS.

Training and mentorship



The Ateneo de Manila University



Liane Peña-Alampay



University of Macau



Shu Fai Cheung



PennState.

The Pennsylvania State University



Sy-Miin Chow



Michael A.
Russell

My work integrates **developmental science, quantitative methods, and dynamic modeling** to study mechanisms and heterogeneity in change across development, family life, and health—with **rigorous, usable methods**.

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Addressing Mechanisms and Heterogeneity in Change with Usable Tools

Training and mentorship

- Before I get into the studies, I want to briefly situate the kind of work I do and how I was trained for it.
- At Ateneo de Manila University, I built my foundation in **psychology**, with strong grounding in *development, context, and teaching*. My mentor there, Liane Peña-Alampay, is a Penn State HDFS alumna, so that connection makes being here especially meaningful to me.
- At Ateneo, I also had the chance to **teach courses** in statistics, psychometrics, and psychology. That experience shaped how I think about methods: not as ends in themselves, but as tools that should be explainable, useful, and connected to substantive questions.
- At the University of Macau, working with Shu Fai Cheung, I deepened my training in **structural equation modeling, meta-analysis, and statistical computation**.
- Then here at Penn State, through PAMT and my work with Sy-Miin Chow and Michael Russell, I extended that foundation into **dynamic models and intensive longitudinal data**, especially in applications tied to *health and behavior* in everyday life.

Training and mentorship



My work integrates **developmental science, quantitative methods, and dynamic modeling** to study mechanisms and heterogeneity in change across development, family life, and health—with **rigorous, usable methods**.

An intuition for inertia



Inertia: how much the past carries forward into the present.

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with Usable Tools

└─ An intuition for inertia

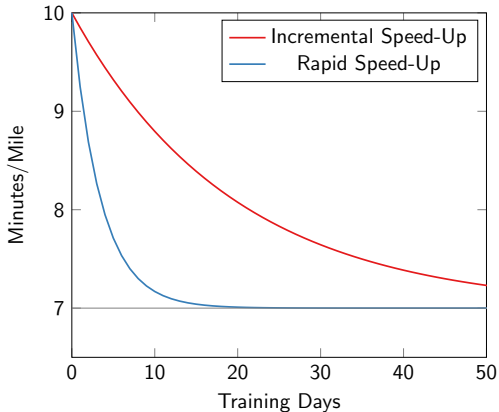
An intuition for inertia



Inertia: how much the past carries forward into the present.

- The first intuition I want to put on the table is inertia.
- A lot of the processes we care about in HDFS **do not reset from one moment to the next**. What happened yesterday, or even a few hours ago, can still shape what happens today.
- That basic idea is what I mean by **inertia**: how much the past carries forward into the present.

Inertia changes trajectories

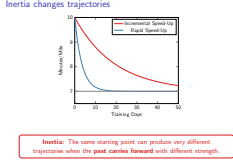


Inertia: The same starting point can produce very different trajectories when the **past carries forward** with different strength.

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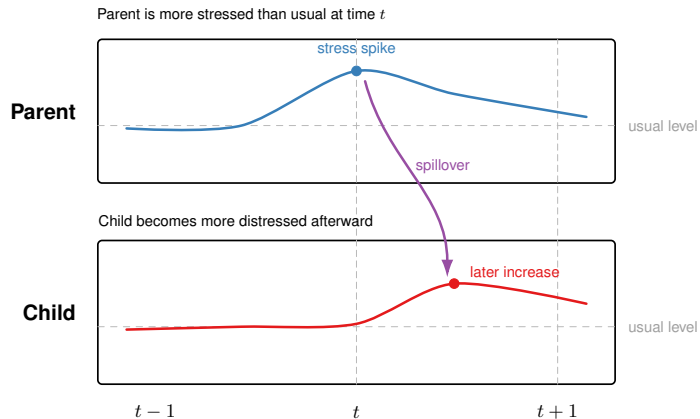
Inertia changes trajectories



- Here both runners start at the same place, but their trajectories diverge over time.
- The difference is not where they began. The difference is how strongly earlier states carry forward.
- When inertia is stronger, the past keeps exerting influence for longer, so the trajectory changes more slowly.
- When inertia is weaker, the system adjusts more quickly and settles back faster.
- That is the key intuition I want to carry into the rest of the talk: dynamic processes are shaped not only by level, but by persistence.
- In HDFS, that kind of persistence shows up in affective recovery, stress carryover, sleep, health behavior, and many other daily-life processes.
- But dynamic processes are not only self-persistent. They can also shape one another.

Coupling and spillover

Escalation / Spillover

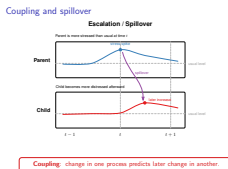


Coupling: change in one process predicts later change in another.

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Coupling and spillover



- That brings us to coupling.
- Here, a parent's stress spike is followed by greater child distress. So now we move from persistence within one process to spillover across connected processes.
- This is the second core idea for the talk: dynamic systems are not just about how a variable carries itself forward, but also about how change in one process predicts later change in another.
- Those two ideas—**inertia** and **coupling**—set up the three studies.
- In Study 1, I ask how direct and indirect effects unfold across time.
- In Study 2, I look at how different dynamic patterns emerge across people and situations.
- And in Study 3, I ask how person-specific dynamics can be synthesized into population-level inference.
- So with that intuition in place, let me start with Study 1, where the central question is how dynamic effects propagate across time through mediating processes.



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Inferences and Effect Sizes for Direct, Indirect, and Total Effects in Continuous-Time Mediation Models

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²Department of Biobehavioral Health, The Pennsylvania State University

³Department of Human Development and Family Studies, The Pennsylvania State University

Question: How do mediation effects unfold across time when the process is modeled continuously rather than at one arbitrary lag?

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Addressing Mechanisms and Heterogeneity in Change with Stable Tools

Study 1

- Once we allow processes to persist and to influence one another, a natural next question is whether one process affects another through an intermediate pathway—and how that unfolds across time.
- So I will start with Study 1, where I ask how mediation effects unfold when we model the process in continuous time instead of tying it to one arbitrary lag.



Why mediation has been a recurring part of my work

- ▶ **Substantive** (e.g.; Bernardo, Wang, Pesigan, & Yeung, 2017; Pesigan, Luyckx, & Alampay, 2014; Zhang, Ku, Wu, Yu, & Pesigan, 2020)
- ▶ **Methodological** (e.g.; Cheung & Pesigan, 2023b; Cheung, Pesigan, & Vong, 2023; Pesigan & Cheung, 2020, 2024; Pesigan, Sun, & Cheung, 2023).
 - ▶ <https://CRAN.R-project.org/package=semccci>
 - ▶ <https://CRAN.R-project.org/package=semblci>
 - ▶ <https://CRAN.R-project.org/package=betaDelta>
 - ▶ <https://CRAN.R-project.org/package=betaDSandwich>

Study 1 extends that line of work into continuous time by adding **standardized effect sizes**, principled **uncertainty quantification**, and an open-source tool in cTMed.

Addressing Mechanisms and Heterogeneity in Change with Stable Tools

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Why mediation has been a recurring part of my work

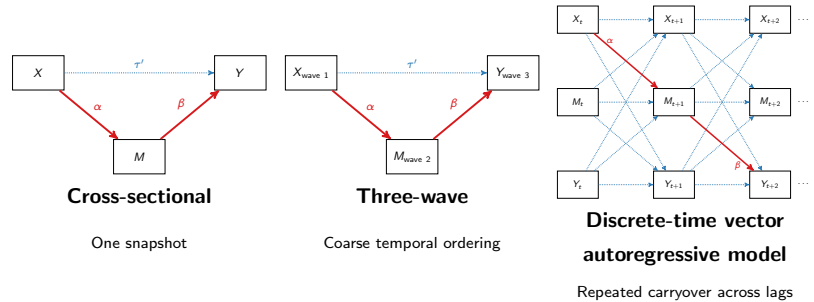
- Mediation has been a recurring part of my work from both substantive and methodological angles.
- On the substantive side, I have used mediation to study mechanisms in applied problems, including alcohol use and related health behaviors.
- On the methods side, I have worked on inference for indirect effects and related quantities, including confidence intervals, bootstrap and Monte Carlo approaches, and standardized effect-size inference.
- Study 1 extends that line of work into continuous time.
- Prior work in continuous-time mediation showed how direct, indirect, and total effects unfold across time. My contribution was to add standardized effect sizes, principled uncertainty quantification, and an open-source implementation that researchers can actually use.

Why mediation has been a recurring part of my work

- ▶ **Substantive** (e.g.; Bernardo, Wang, Pesigan, & Yeung, 2017; Pesigan, Luyckx, & Alampay, 2014; Zhang, Ku, Wu, Yu, & Pesigan, 2020)
- ▶ **Methodological** (e.g.; Cheung & Pesigan, 2023b; Cheung, Pesigan, & Vong, 2023; Pesigan & Cheung, 2020, 2024; Pesigan, Sun, & Cheung, 2023).
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Study 1 extends that line of work into continuous time by adding **standardized effect sizes**, principled **uncertainty quantification**, and an open-source tool in cTMed.

Why this problem matters



- ▶ Mediation is fundamentally a **temporal process**.
- ▶ But in discrete time, the size of the effect depends on the **chosen measurement interval**.

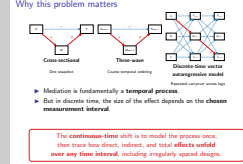
The **continuous-time** shift is to model the process once, then trace how direct, indirect, and total **effects unfold over any time interval**, including irregularly spaced designs.

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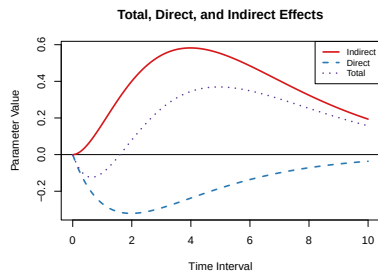
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Why this problem matters

- This slide gets at why the problem matters.
- Mediation is fundamentally a temporal process, but many of our traditional models give us only a few snapshots of that process.
- Cross-sectional models give us one snapshot, three-wave models give us coarse temporal ordering, and even discrete-time autoregressive models still depend on the particular measurement interval we happened to choose.
- The key shift in continuous time is that we model the underlying process once, and then trace how direct, indirect, and total effects unfold over any time interval.
- That means we can study mediation as a genuinely dynamic process, including in designs with irregular spacing.
- For HDFS researchers, the payoff is that mechanisms in development, family process, and health behavior rarely operate at one arbitrary lag.



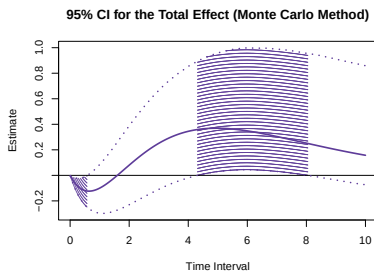
Novel contributions



Effect size measures

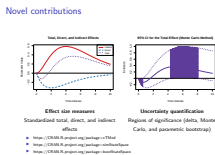
Standardized total, direct, and indirect effects

- ▶ <https://CRAN.R-project.org/package=cTMed>
- ▶ <https://CRAN.R-project.org/package=simStateSpace>
- ▶ <https://CRAN.R-project.org/package=bootStateSpace>



Uncertainty quantification

Regions of significance (delta, Monte Carlo, and parametric bootstrap)



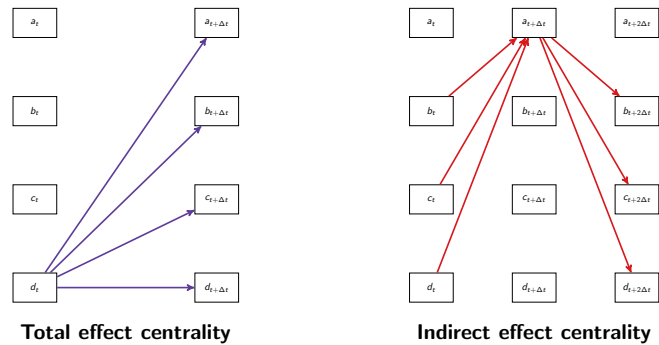
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Addressing Mechanisms and Heterogeneity in Change with Suitable Tools

- Novel contributions

- This slide highlights the three main contributions of Study 1, using a simple example: **stress** affecting **mood** through **coping**.
- The key idea is that in continuous-time mediation, we ask not just whether coping mediates the effect of stress on mood, but **how that mediation unfolds across time**.
- Right after a stressful event, the effect on mood may be mostly direct; a little later, coping may begin to carry more of that effect.
- So the direct, indirect, and total effects are **time-dependent**, not fixed.
- My first contribution is introducing **standardized** total, direct, and indirect effects.
- My second is **uncertainty quantification** across time using delta, Monte Carlo, and parametric bootstrap methods.
- My third is implementing these ideas in the open-source cTMed package.
- So the broader payoff is a practical way to study **when an effect is immediate, when it becomes mediated, and how it evolves over time**.

One direction this line of work opens up



Which variable matters most for shaping the rest of the system over time?

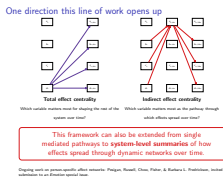
Which variable matters most as the pathway through which effects spread over time?

This framework can also be extended from single mediated pathways to **system-level summaries** of how effects spread through dynamic networks over time.

Ongoing work on person-specific affect networks: Pesigan, Russell, Chow, Fisher, & Barbara L. Fredrickson, invited submission to an *Emotion* special issue.

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One direction this line of work opens up



- One reason I am excited about this work is that it opens up a broader systems perspective.
- Study 1 focused on mediation one pathway at a time, but many substantive questions are really about how an **entire affective system reorganizes** over time.
- That is the idea behind an ongoing project on **person-specific affect networks during meditation training**, across baseline, intervention, and follow-up.
- Here, **total-effect centrality** asks which emotion has the *broadest downstream* influence, and **indirect-effect centrality** asks which emotion serves as a key *conduit* through which effects spread.
- Because the model is continuous-time, we can examine those roles across **multiple timescales**—from short-term ripple effects to longer-horizon accumulation.
- And because this is an intervention study, we can link that reorganization to **practice dose** and to **follow-up affective functioning and well-being**.
- So the broader payoff of Study 1 is a framework for turning complex dynamics into interpretable summaries that show **which variables have the widest downstream influence and which variables serve as the main pathways through which effects spread over time**.
- Study 2 then shifts to a related question: how qualitatively different dynamic patterns emerge across people.



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Common and Unique Latent Transition Analysis (CULTA) as a Way to Examine the Trait–State Dynamics of Alcohol Intoxication

Ivan Jacob Agaloos Pesigan¹, Michael A. Russell^{1, 2}, and Sy-Miin Chow³

¹ Edna Bennett Pierce Prevention Research Center, College of Health and Human Development, The Pennsylvania State University

² Department of Biobehavioral Health, The Pennsylvania State University

³ Department of Human Development and Family Studies, The Pennsylvania State University

Question: Can inertia help distinguish qualitatively different drinking patterns, beyond differences in average intoxication level?

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Addressing Mechanisms and Heterogeneity in Change with Study 2

Study 2

- Study 1 focused on how dynamic effects unfold across time.
- Study 2 asks a different but closely related question: can inertia itself help define qualitatively different risk patterns, beyond differences in average intoxication?
- This is where I move from quantifying dynamic effects to building a new latent-variable framework for decomposing heterogeneity in intensive longitudinal alcohol data.

Study 2



Question: Can inertia help distinguish qualitatively different drinking patterns, beyond differences in average intoxication level?

Why this substantive problem demanded a new model

Why Study 2 is exciting

- ▶ **Substantive grounding:** Alcohol and addiction have been a consistent part of my applied work (e.g. Yu, Pesigan, Zhang, & Wu, 2019; Yu, Wu, & Pesigan, 2015; Zamboanga, Iwamoto, Pesigan, & Tomaso, 2015; Zamboanga, Pesigan, Tomaso, Schwartz, et al., 2015; Zhang, Ku, Wu, Yu, & Pesigan, 2020; Zhang, Pesigan, Kahler, Yip, et al., 2019).
- ▶ **Substantive question:** How do risky drinking patterns persist or change across days?
- ▶ **Methodological challenge:** Transdermal alcohol concentration (TAC) features are multidimensional, partly shared, partly feature-specific, and dynamic over time.

Substantive grounding in addiction research, together with the quantitative expertise to build **new models** when the science demands them.

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Addressing Mechanisms and Heterogeneity in Change with Study 2

Why this substantive problem demanded a new model

- This study is a natural next step for me.
- Addiction and alcohol have been a consistent substantive thread in my work, but Study 2 is where that background meets the kind of quantitative contribution I want to make in HDFS.
- TAC sensors give us richer and more objective data, but they also create a harder modeling problem: the features are partly shared, partly feature-specific, and changing across days.
- That is exactly the setting that motivated CULTA.

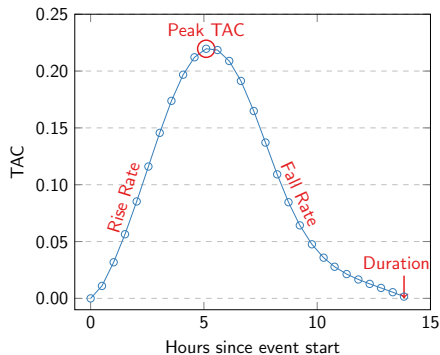
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Substantive grounding in addiction research, together with the quantitative expertise to build new models when the science demands them.

From TAC curves to dynamic questions



Simple summaries of the TAC features miss three things:

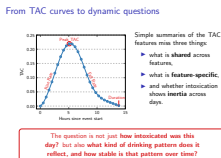
- ▶ what is **shared** across features,
- ▶ what is **feature-specific**,
- ▶ and whether intoxication shows **inertia** across days.

The question is not just **how intoxicated was this day?** but also **what kind of drinking pattern does it reflect, and how stable is that pattern over time?**

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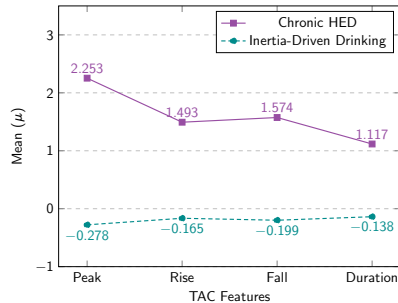
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From TAC curves to dynamic questions

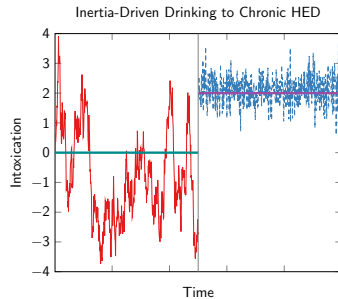


- Here, the key data are **repeated day-specific TAC curves**: across the monitoring period, each participant can contribute multiple drinking-day curves, and from each curve we extract features like peak, rise rate, fall rate, and duration.
- But once we do that, a new question appears: are we looking at one common intoxication process, feature-specific expressions of intoxication, or both?
- And beyond that, does intoxication carry over across days?
- So the substantive question becomes not just how intoxicated this day was, but what kind of drinking pattern it reflects and how stable that pattern is over time.
- That is exactly the problem CULTA was designed to solve.

Profiles differed in mean intoxication and inertia



Difference in mean levels



Difference in temporal persistence

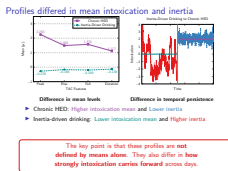
- ▶ Chronic HED: Higher intoxication mean and Lower inertia
- ▶ Inertia-driven drinking: Lower intoxication mean and Higher inertia

The key point is that these profiles are **not defined by means alone**. They also differ in **how strongly intoxication carries forward** across days.

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Addressing Mechanisms and Heterogeneity in Change with Stable Tools

Profiles differed in mean intoxication and inertia



- CULTA answers that question by separating overall intoxication level from temporal persistence.
- The main empirical result is that two profiles emerged, but the key point is how they differ.
- Chronic HED is characterized by higher mean intoxication and lower inertia, while inertia-driven drinking shows lower mean intoxication but stronger carryover across days.
- So this is not just a classification result. It shows that risky drinking patterns differ not only in level, but also in temporal persistence.

AUD risk predicts getting stuck in chronic HED

Start in chronic HED	Stay in chronic HED	Shift into chronic HED
No risk → Risk → Likely AUD	No risk → Risk → Likely AUD	No risk → Risk → Likely AUD
0.03 → 0.07 → 0.15	0.21 → 0.30 → 0.40	0.03 → 0.06 → 0.10

CULTA does not just classify drinking days; it helps identify who is likely to get stuck in riskier dynamic patterns.

Addressing Mechanisms and Heterogeneity in Change with Stable Tools

AUD risk predicts getting stuck in chronic HED

- AUD risk helps identify who is more likely to start in, remain in, or shift into the chronic HED profile.
- The main pattern is simple: as AUD risk increases, so does the probability of getting stuck in the chronic pattern.
- So Study 2 shows that risky drinking patterns are not defined by average intoxication alone, but also by persistence and who gets stuck in maladaptive dynamics over time.
- And once we start treating these dynamic signatures as meaningful individual differences, the next question is how to synthesize them at the population level without averaging them away.
- That is exactly the problem I take up in Study 3.

AUD risk predicts getting stuck in chronic HED

Start in chronic HED	Stay in chronic HED	Shift into chronic HED
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CULTA does not just classify drinking days; it helps identify who is likely to get stuck in riskier dynamic patterns.

Under review at *Multivariate Behavioral Research*

From Person-Specific Dynamics to Population Inference: A Computationally Efficient Meta-Analytic Structural Equation Modeling Framework for Integrating Complex Multilevel Dynamic Models

Ivan Jacob Agalloos Pesigan¹, Michael A. Russell^{1,2}, and Sy-Miin Chow³

¹Edna Bennett Pierce Prevention Research Center, The Pennsylvania State University

²Department of Biobehavioral Health, The Pennsylvania State University

³Department of Human Development and Family Studies, The Pennsylvania State University

Question: How can we draw **population-level** conclusions from **person-specific dynamics** without averaging away the very individual differences we care about?

- Study 3 asks how to preserve **person-specific dynamics** while still making valid **population-level** inferences.
- For me, this is not only a statistical problem.
- It is the problem behind many HDFS questions, because we often care about **dynamic parameters** that capture regulation, adaptation, spillover, and risk in ways that **differ meaningfully across people, dyads, and families**.
- So the question is how to synthesize those individual dynamic signatures without averaging away the heterogeneity that makes them substantively important.

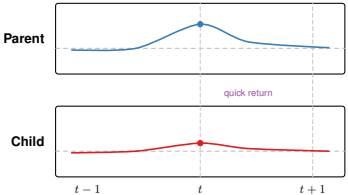
Dyadic phenotypes

Dynamic Phenotype Profiles of Dyadic Functioning

Illustrative parent-child dynamic signatures

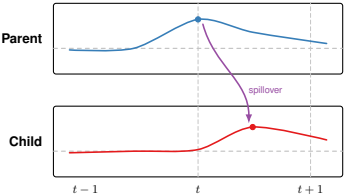
Resilient

brief disturbance, fast recovery



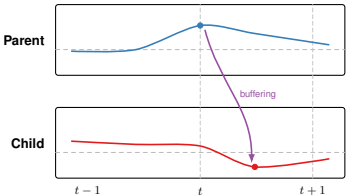
Stress-Contagious

parent distress spills into child distress



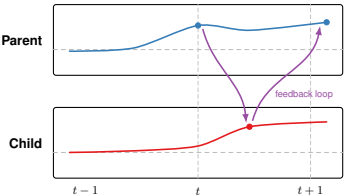
Buffering

parent regulation helps child settle



Volatile

mutual amplification, slow settling

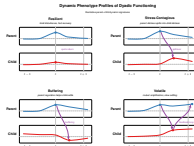


Addressing Mechanisms and Heterogeneity in Change with StudyB

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Dyadic phenotypes

Dyadic phenotypes

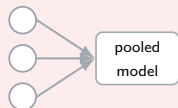


- Before I show the mechanics of MetaVAR, I want to show the kind of scientific target I care about.
- Imagine estimating a dynamic signature for each parent-child dyad: some dyads recover quickly after disturbance, some show stress contagion, some show buffering, and some show mutual amplification.
- Those are not just abstract parameters. They are candidate dynamic phenotypes of dyadic regulation and functioning.
- And once we can estimate those signatures well, the next question is **how to synthesize them without flattening the very heterogeneity that makes them interesting.**
- That is the problem MetaVAR is designed to solve.

The trade-off MetaVAR is designed to solve

Population inference

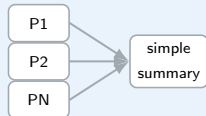
One-step hierarchical models



- + strong for average dynamics
- partial pooling can shrink person-specific dynamics

Idiographic fidelity

Naive two-step workflow

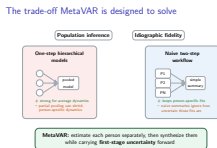


- + keeps person-specific fits
- naive summaries ignore how uncertain those fits are

MetaVAR: estimate each person separately, then synthesize them while carrying **first-stage uncertainty** forward

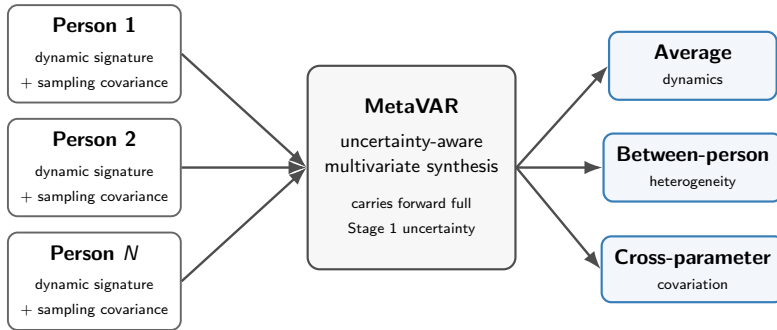
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The trade-off MetaVAR is designed to solve



- There is a real trade-off here.
- One-step hierarchical models are attractive because they estimate population averages and individual differences together. But the cost is that person-specific dynamics can get pulled toward the population mean.
- At the other extreme, we can fit a separate model per person and summarize afterward. That preserves the idiographic fits and scales well, but it can treat all person-level estimates as if they were equally precise.
- MetaVAR is the middle ground: keep the person-specific models, but carry their uncertainty into the second stage.
- In HDFS terms, this matters whenever we care not only about the average person, dyad, or family, but also about how much meaningful variation there is around that average and what predicts it.

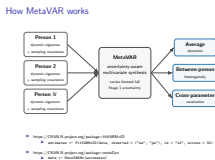
How MetaVAR works



- ▶ <https://CRAN.R-project.org/package=fitVARMxID>
 - ▶ `estimates <- FitVARMxID(data, observed = ("na", "pa"), id = "id", ncores = 32)`
- ▶ <https://CRAN.R-project.org/package=metaDyn>
 - ▶ `meta <- MetaVARMx(estimates)`

Addressing Mechanisms and Heterogeneity in Change with Sturdy Tools

└ How MetaVAR works



- Stage 1 is person-specific estimation: each individual gets an idiographic model, which yields a dynamic signature and its sampling covariance matrix.
- Stage 2 is MetaVAR: instead of treating those person-specific estimates as error-free, the method synthesizes both the signatures and their uncertainty.
- That gives us valid population-level inference on average dynamics, between-person heterogeneity, and covariation among dynamic parameters.

Simulation results: MetaVAR is strongest for heterogeneity

Coverage heatmap by parameter, design cell, and method

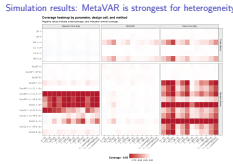
Negative values indicate undercoverage; zero indicates nominal coverage.



Addressing Mechanisms and Heterogeneity in Change with Study Tools

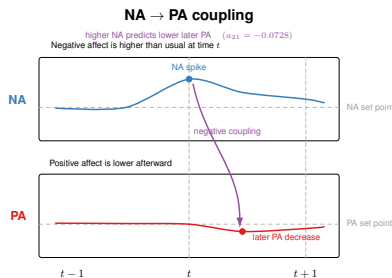
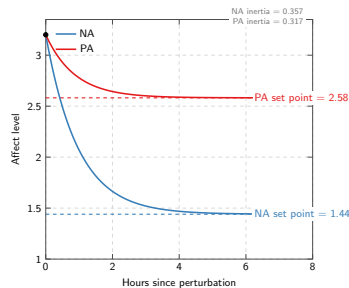
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Simulation results: MetaVAR is strongest for heterogeneity



- The simulation clarifies where MetaVAR is especially useful.
- Its clearest advantage is when the scientific target is **heterogeneity**: how much people differ, and how well we calibrate uncertainty about those differences.
- For population averages, the Bayesian multilevel comparator remained competitive.
- So I see MetaVAR not as a universal replacement, but as a strong option when we want to preserve idiographic estimation while still drawing valid population-level inferences about heterogeneity.

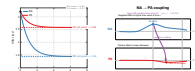
Empirical illustration: Average affect dynamics



Addressing Mechanisms and Heterogeneity in Change with StudyB Tools

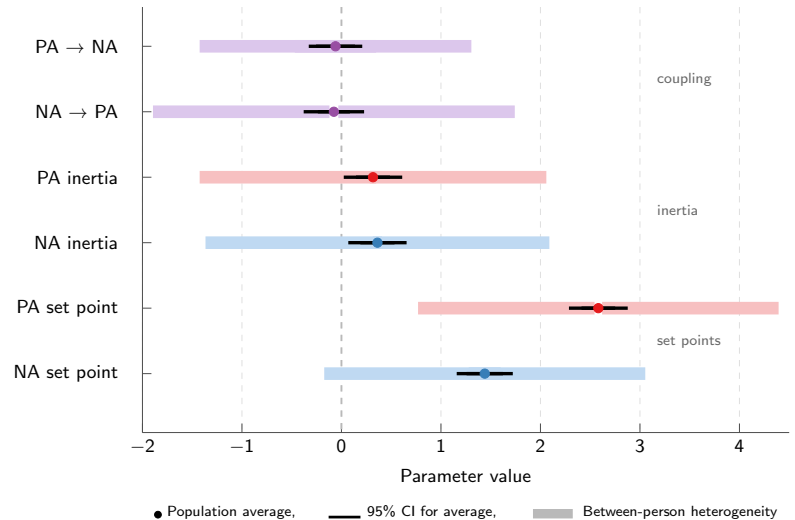
Empirical illustration: Average affect dynamics

Empirical illustration: Average affect dynamics



- The average affect system is interpretable: on average, people show higher positive affect than negative affect, and both processes show positive inertia.
- I am also showing one representative cross-lagged result here: higher negative affect predicts lower later positive affect.
- The companion result goes in the same direction for positive affect predicting lower later negative affect, so the average system reflects reciprocal inhibitory coupling.
- So MetaVAR gives us an interpretable population-level affect system, not just a collection of person-specific estimates.

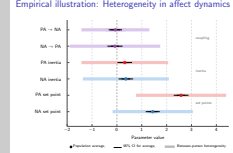
Empirical illustration: Heterogeneity in affect dynamics



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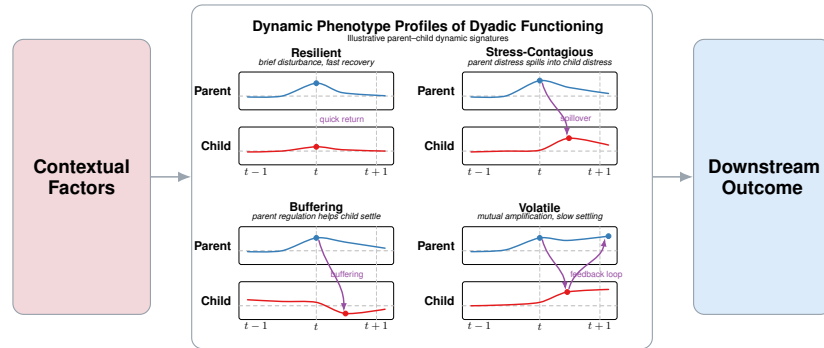
Addressing Mechanisms and Heterogeneity in Change with StudyB Tools

Empirical illustration: Heterogeneity in affect dynamics



- The more important result is not only where the average lies, but how much people differ around it.
- The dot is the population average, the thin interval is the 95% confidence interval for that average, and the wider band shows approximate between-person heterogeneity around it.
- That wider band comes from the random-effects variance for each parameter, so it reflects population spread rather than uncertainty about the mean.
- So people differ meaningfully in their set points, in how persistent affect is, and in how strongly negative and positive affect influence one another over time.
- The payoff is that heterogeneity stops being a nuisance term and becomes a substantive result: people differ in their set points, persistence, coupling, and co-fluctuation, and those differences are exactly what we often care about in HDFS.

Examining heterogeneity and downstream effects



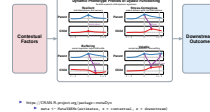
- ▶ <https://CRAN.R-project.org/package=metaDyn>

```
► meta <- MetaVARmX(estimates, x = contextual, z = downstream)
```

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Addressing Mechanisms and Heterogeneity in Change with Stusys Tools

Examining heterogeneity and downstream effects



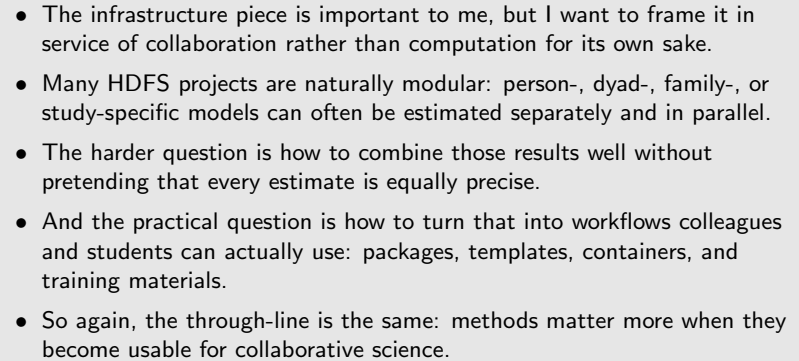
- This slide shows why MetaVAR matters beyond a single methods paper.
- In a broader SEM framework, person-specific dynamic signatures can be modeled as outcomes of contextual factors and as predictors of later outcomes.
- So instead of stopping at “people differ,” we can ask what predicts those differences and why they matter downstream.
- That is the bridge to the next slide: toward dynamic phenotypes in HDFS.

Future directions: Making dynamic synthesis usable

The diagram illustrates a three-stage process for making dynamic synthesis usable:

- Fast local models at scale:** This stage involves running many local models in parallel using various methods (e.g., linear, global, local, or model-specific approaches) to generate evidence.
 - Methods: linear, global, local, or model-specific approaches
- Synthesizing evidence rigorously:** This stage involves combining evidence from multiple models using a central synthesis engine (e.g., model averaging, model stacking, or model selection) to produce a single, more accurate prediction.
 - Methods: model averaging, model stacking, or model selection
- Delivering usable results:** This stage involves presenting the synthesized results in a user-friendly format (e.g., interactive dashboards and reports) to support decision-making.
 - Methods: interactive dashboards and reports, open tools, synthetic confidence, and decision audit systems

Arrows indicate the flow from left to right, showing the progression from local models to evidence synthesis and finally to usable results.



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Addressing Mechanisms and Heterogeneity in Change with Usable Tools

Matri meae carissimae, VAP.

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Addressing Mechanisms and Heterogeneity in Change with Usable Tools

Ivan Jacob Agaloos Pesigan (Jek)
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Postdoctoral Fellow
Prevention and Methodology Training Program (PAMT)
Edna Bennett Pierce Prevention Research Center
The Pennsylvania State University

April 20, 2026



PennState

<https://github.com/jeksterslab/talks20260420PSUHDFSJobtalk>

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- Thank you very much for your attention.
- I have tried to show a research program centered on dynamic processes: how change unfolds over time, how people and relationships differ in those processes, and how we can build tools that make those questions easier to study well.
- Across the three studies, the common thread is not methodology for its own sake, but usable quantitative tools for substantive questions in development, family life, health, and daily contexts.
- What excites me most about HDFS is that these are exactly the kinds of questions the field asks.
- I would be excited to contribute here through methods development, collaboration, student training, and building usable quantitative infrastructure that helps substantive researchers ask better questions about development, family life, and health.
- Thank you, and I would be very happy to take questions

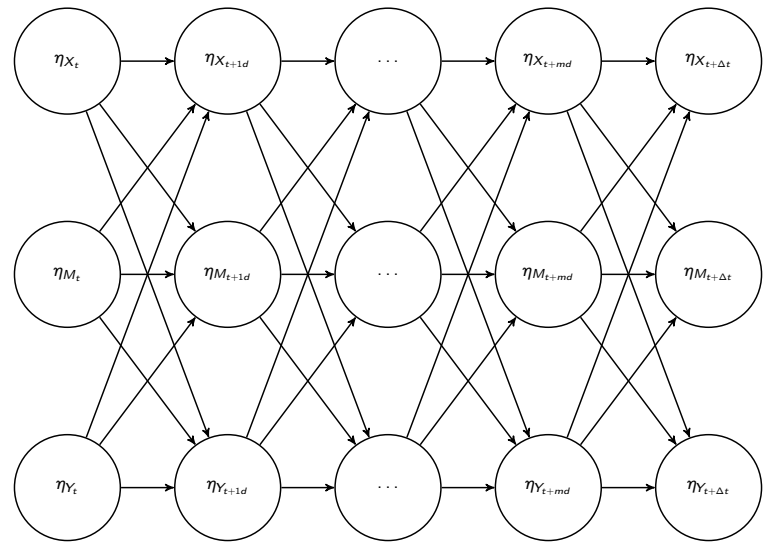
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Addressing Mechanisms and Heterogeneity in Change
with Usable Tools
└─ Extra slides

[Extra slides](#)

Extra slides

Continuous-Time Vector Autoregressive Model (CT-VAR: Drift Matrix Φ)

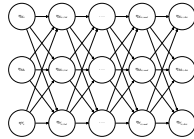


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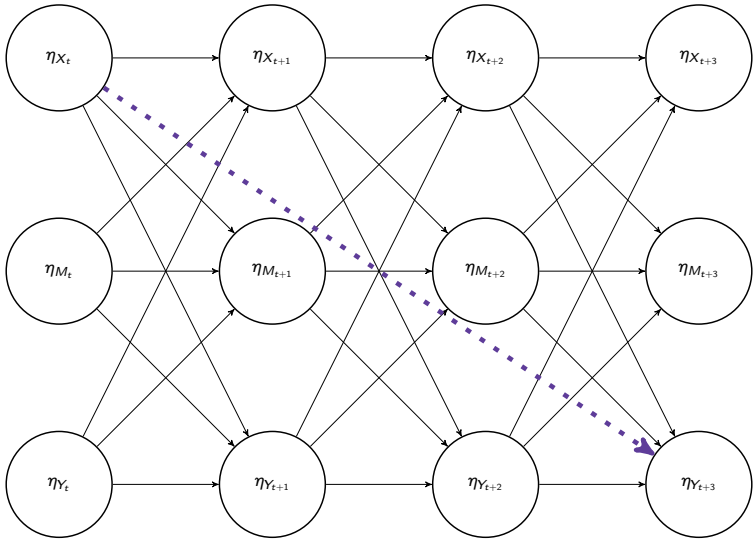
Addressing Mechanisms and Heterogeneity in Change with Usable Tools

- Extra slides
- Continuous-Time Vector Autoregressive Model (CT-VAR: Drift Matrix Φ)

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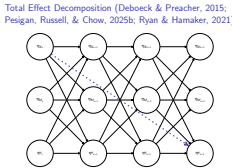
Total Effect Decomposition (Deboeck & Preacher, 2015; Pesigan, Russell, & Chow, 2025b; Ryan & Hamaker, 2021)



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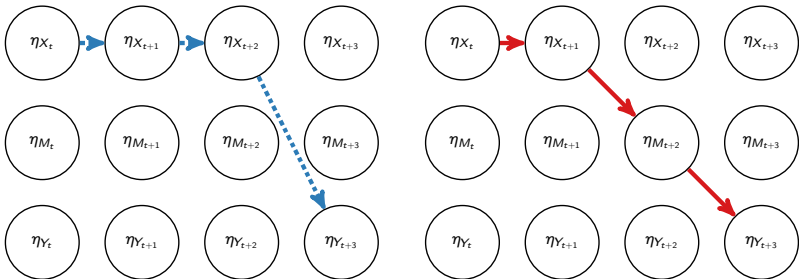
Addressing Mechanisms and Heterogeneity in Change
with Usable Tools

- Extra slides
 - Total Effect Decomposition (Deboeck & Preacher, 2015; Pesigan, Russell, & Chow, 2025b; Ryan & Hamaker, 2021)



Total Effect Decomposition

Blue = direct effect ($\beta_{11}\beta_{11}\beta_{31}$). Red = indirect effect ($\beta_{11}\beta_{21}\beta_{32}$).



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Addressing Mechanisms and Heterogeneity in Change with Usable Tools

└ Extra slides

└ Total Effect Decomposition

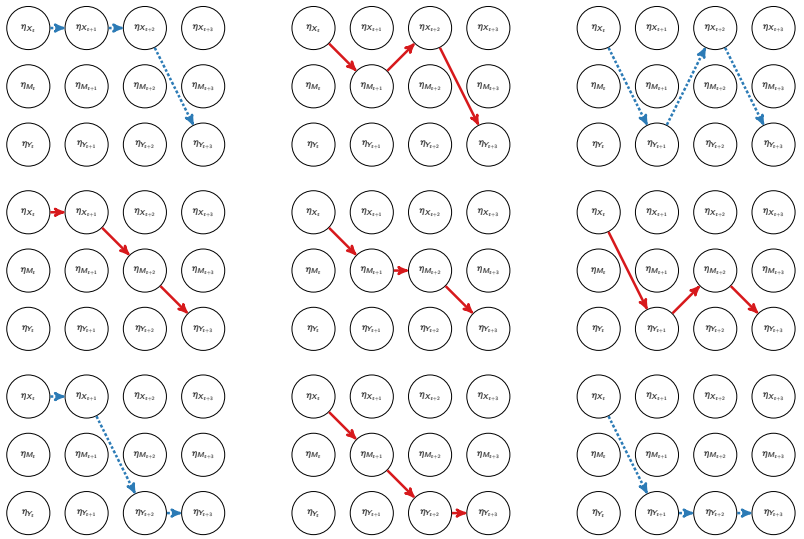
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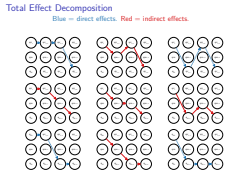
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Addressing Mechanisms and Heterogeneity in Change
with Usable Tools
└ Extra slides
└ Total Effect Decomposition



Total Effect Decomposition

Total, Direct, and Indirect Effects:

$$\text{Total} = \exp(\Delta t \Phi) = \beta_{\Delta t} \tag{1}$$

$$\text{Direct} = \exp(\Delta t \mathbf{D}_m \Phi \mathbf{D}_m) \tag{2}$$

$$\text{Indirect} = \text{Total} - \text{Direct}. \tag{3}$$

Standardized Effects: Effect Size Measures

$$\text{Total}_{i,j}^* = \text{Total}_{i,j} \left(\frac{\sigma_{x_j}}{\sigma_{y_i}} \right), \tag{4}$$

$$\text{Direct}_{i,j}^* = \text{Direct}_{i,j} \left(\frac{\sigma_{x_j}}{\sigma_{y_i}} \right), \quad \text{and} \tag{5}$$

$$\text{Indirect}_{i,j}^* = \text{Total}_{i,j}^* - \text{Direct}_{i,j}^* \tag{6}$$

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Addressing Mechanisms and Heterogeneity in Change
with Usable Tools
└─ Extra slides
└─ Total Effect Decomposition

Total Effect Decomposition

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CT-VAR (Chow et al., 2023; Deboeck & Preacher, 2015)

Measurement component:

$$\mathbf{y}_{i,t_m} = \boldsymbol{\nu} + \mathbf{A}\boldsymbol{\eta}_{i,t_m} + \boldsymbol{\varepsilon}_{i,t_m}, \boldsymbol{\varepsilon}_{i,t_m} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Theta}) \tag{7}$$

Dynamic component:

$$d\boldsymbol{\eta}_{i,t} = (\boldsymbol{\Phi}\boldsymbol{\eta}_{i,t}) dt + \boldsymbol{\Sigma}^{\frac{1}{2}} d\mathbf{W}_{i,t} \tag{8}$$

Three-variable CT-VAR mediation model:

$$\mathbf{y} = \begin{pmatrix} X \\ M \\ Y \end{pmatrix}, \quad \text{and} \quad \boldsymbol{\eta} = \begin{pmatrix} \eta_X \\ \eta_M \\ \eta_Y \end{pmatrix} \tag{9}$$

Integral form:

$$\boldsymbol{\eta}_{i,t_{m_i}} = \boldsymbol{\alpha}_{\Delta t_{m_i}} + \boldsymbol{\beta}_{\Delta t_{m_i}} \boldsymbol{\eta}_{i,t_{m_i}-1} + \boldsymbol{\zeta}_{i,t_{m_i}}, \boldsymbol{\zeta}_{i,t_{m_i}} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Psi}_{\Delta t_{m_i}}) \tag{10}$$

$$\boldsymbol{\beta}_{\Delta t_{m_i}} = \exp(\Delta t \boldsymbol{\Phi}) \tag{11}$$

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Addressing Mechanisms and Heterogeneity in Change with Usable Tools

└ Extra slides

└ CT-VAR (Chow et al., 2023; Deboeck & Preacher, 2015)

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Delta Method Confidence Intervals

We define the delta method following the multivariate central limit theorem. Let θ be $\text{vec}(\Phi)$, that is, the elements of the drift matrix Φ in vector form sorted column-wise. Let $\hat{\theta}$ be $\text{vec}(\hat{\Phi})$, that is, the estimates of the elements of the drift matrix. The uncertainty around any functions of the estimates for $\hat{\theta}$ is assumed to be characterized as:

$$\sqrt{n} \left(\mathbf{g}(\hat{\theta}) - \mathbf{g}(\theta) \right) \xrightarrow{D} \mathcal{N}(0, \mathbf{J}\mathbf{\Gamma}\mathbf{J}') \tag{12}$$

where $\mathbf{g}$ is a vector of functions, $\mathbf{J}$ is the matrix of first-order derivatives of the functions in $\mathbf{g}$ with respect to the elements of θ and $\mathbf{\Gamma}$ is the asymptotic variance-covariance matrix of $\hat{\theta}$. The approximate distribution of $\mathbf{g}(\hat{\theta})$ is given by

$$\mathbf{g}(\hat{\theta}) \approx \mathcal{N}(\mathbf{g}(\theta), n^{-1}\mathbf{J}\mathbf{\Gamma}\mathbf{J}'). \tag{13}$$

When $\mathbf{\Gamma}$ is unknown, by substitution, we can use the estimated sampling variance-covariance matrix of $\hat{\theta}$, that is, $\hat{\mathbf{V}}(\hat{\theta})$ for $n^{-1}\mathbf{\Gamma}$.

Addressing Mechanisms and Heterogeneity in Change with Usable Tools

- Extra slides
- Delta Method Confidence Intervals

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Monte Carlo (MC) Method Confidence Intervals

Based on the asymptotic properties of maximum likelihood estimators, we can assume that they are normally distributed around the population parameters

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Using this distributional assumption, a sampling distribution of $\hat{\theta}$ which we refer to as $\hat{\theta}^*$ can be generated by replacing the population parameters with sample estimates, that is,

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A sampling distribution of $\mathbf{g}\left(\hat{\theta}\right)$, which we refer to as $\mathbf{g}\left(\hat{\theta}^*\right)$, can be generated by using the simulated estimates to calculate $\mathbf{g}$. The standard deviations of the simulated estimates are the SEs. In resampling approaches, it is customary to use the percentiles of the generated distribution as CIs instead of the SEs.

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Addressing Mechanisms and Heterogeneity in Change with Usable Tools

- └ Extra slides
- └ Monte Carlo (MC) Method Confidence Intervals

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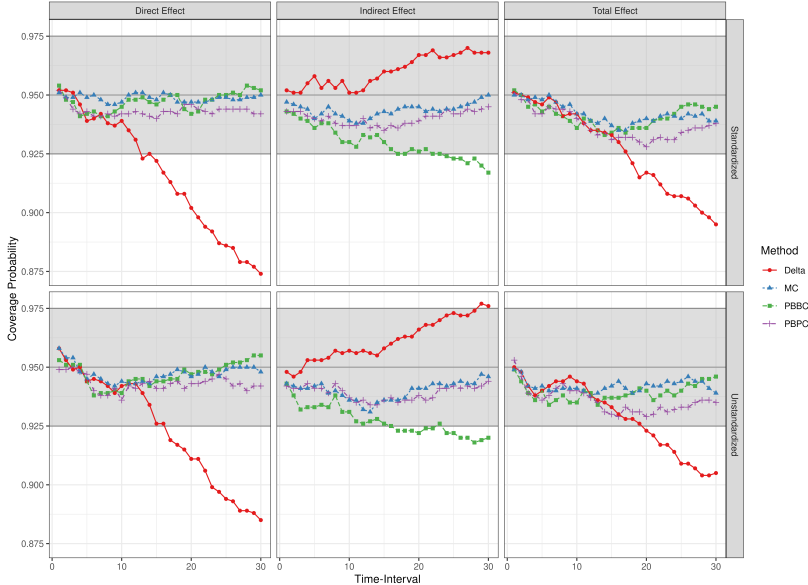
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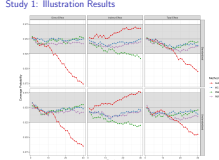
Study 1: Illustration Results



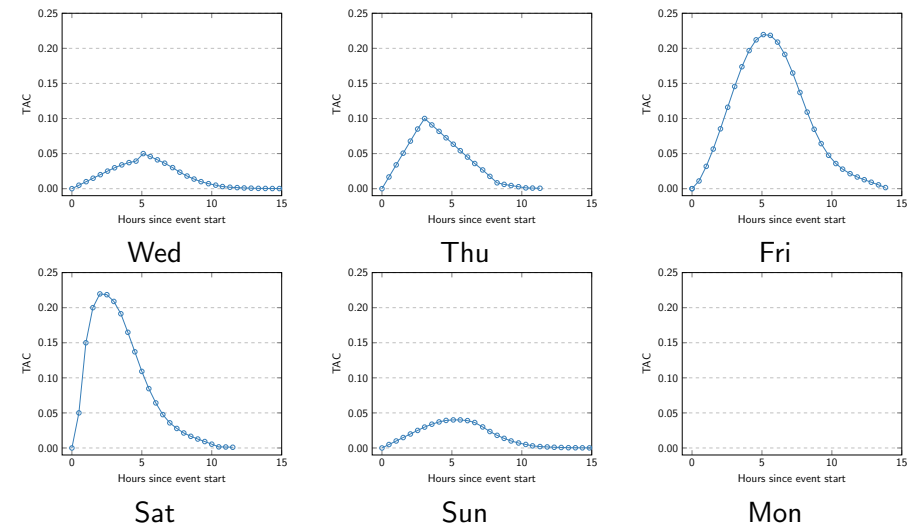
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Addressing Mechanisms and Heterogeneity in Change with Usable Tools

- Extra slides
- Study 1: Illustration Results



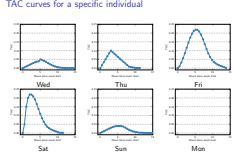
TAC curves for a specific individual



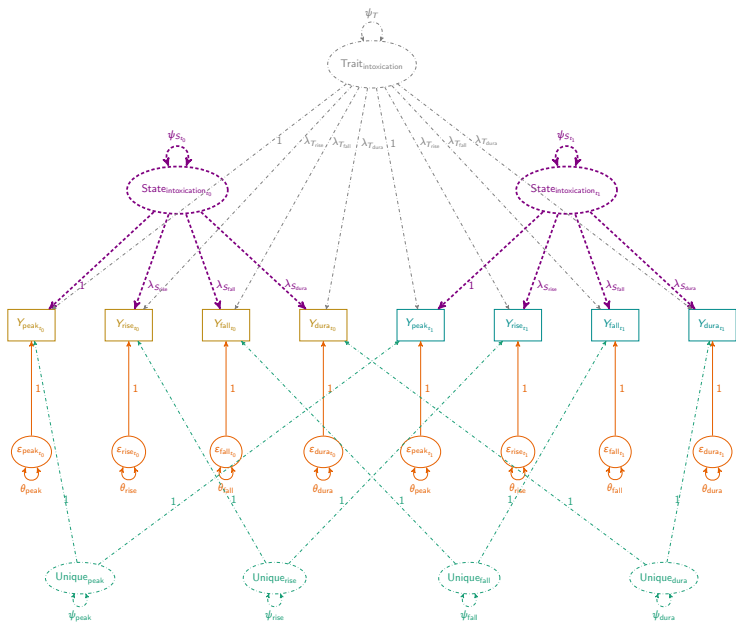
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Addressing Mechanisms and Heterogeneity in Change with Usable Tools

- Extra slides
- TAC curves for a specific individual



CUTS component of the CULTA model

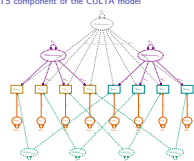


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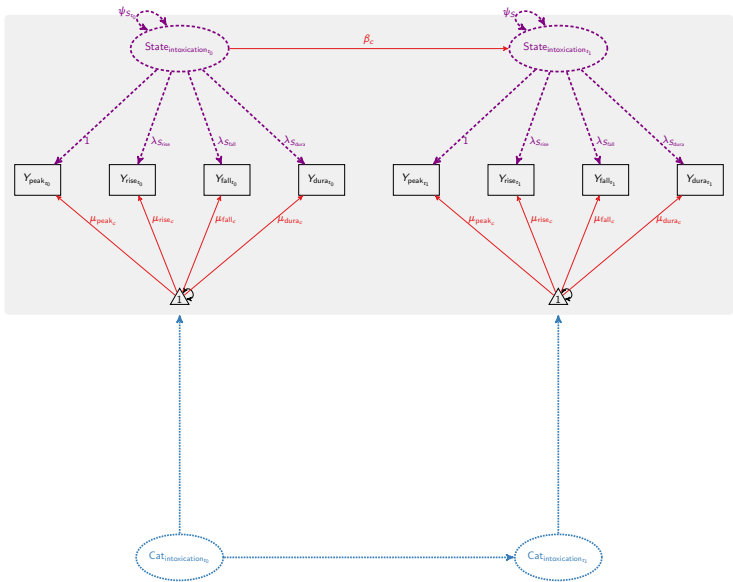
Addressing Mechanisms and Heterogeneity in Change with Usable Tools

- Extra slides
 - CUTS component of the CULTA model

CUTS component of the CULTA model



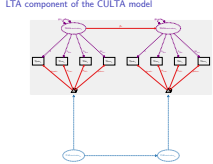
LTA component of the CULTA model



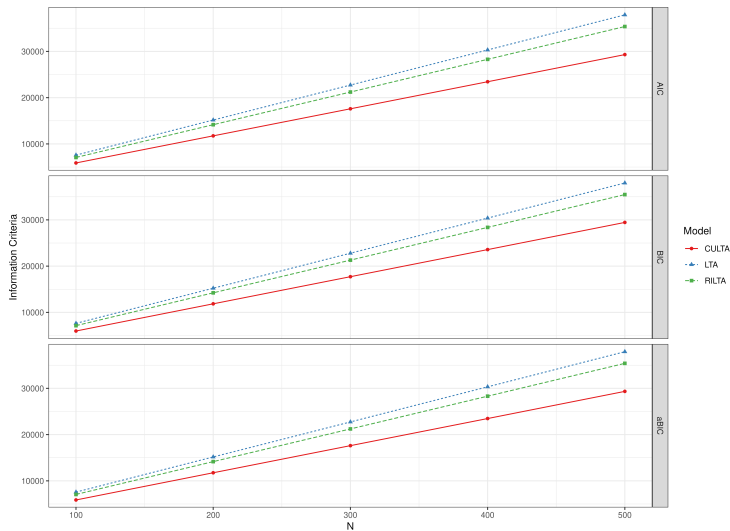
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Addressing Mechanisms and Heterogeneity in Change with Usable Tools

- Extra slides
- LTA component of the CULTA model



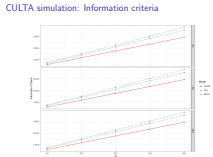
CULTA simulation: Information criteria



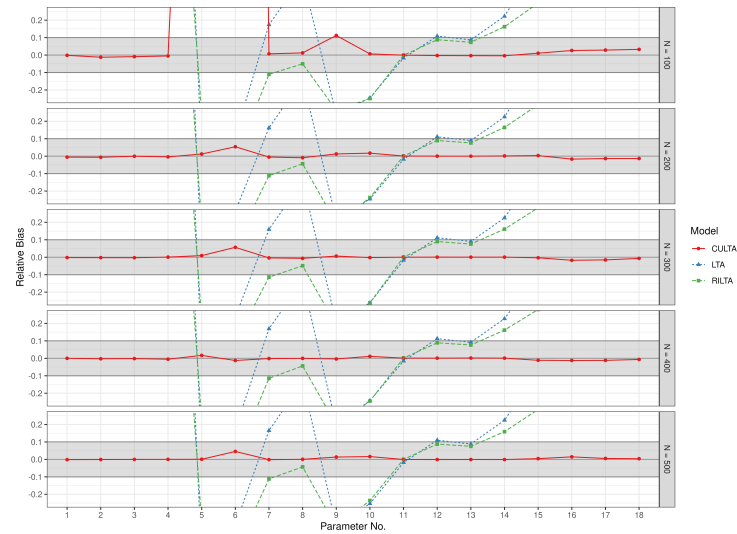
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Addressing Mechanisms and Heterogeneity in Change with Usable Tools

- Extra slides
- CULTA simulation: Information criteria



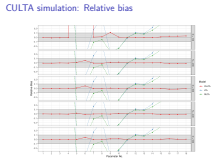
CULTA simulation: Relative bias



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Addressing Mechanisms and Heterogeneity in Change with Usable Tools

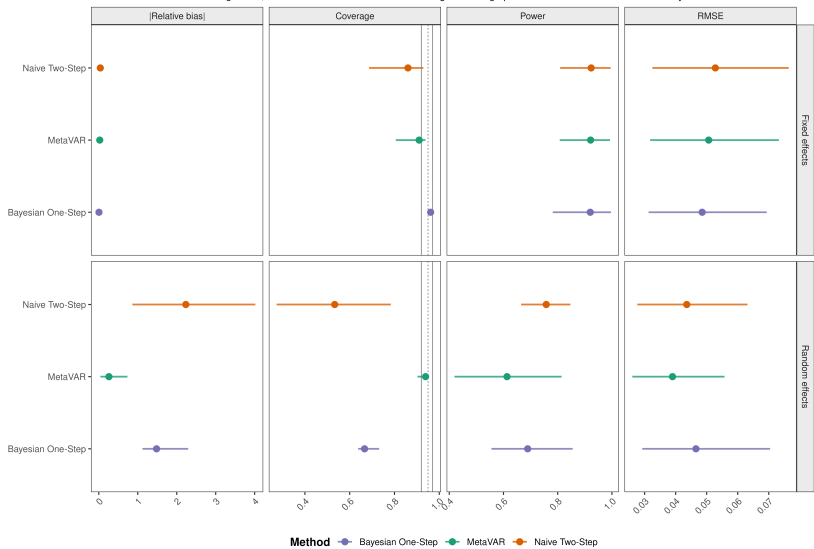
- Extra slides
- CULTA simulation: Relative bias



Simulation results: MetaVAR is strongest for heterogeneity

Overall method performance across simulation design cells

Points are means across design cells; horizontal lines show the min-max range. Coverage panels include nominal .95 and Bradley's liberal .92-.97 band.

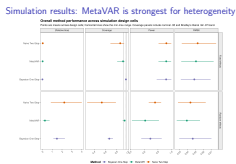


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Addressing Mechanisms and Heterogeneity in Change with Usable Tools

Extra slides

Simulation results: MetaVAR is strongest for heterogeneity

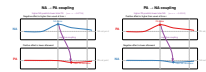


- The simulation tells me where MetaVAR is most useful. I do not present it as a universal replacement for one-step models. For population averages, the Bayesian multilevel comparator stayed competitive. MetaVAR's clearest advantage was when the scientific target was heterogeneity: how much people differ, and how well we calibrate uncertainty about those differences.

2026-04-19

- Extra slides

Empirical illustration: Coupling



NA

Negative affect is higher than usual at time t



PA

 $t-1$ t $t + 1$

PA

Positive affect is higher than usual at time t .



NA

 $t-1$

t

 $t + 1$

2026-04-19

Addressing Mechanisms and Heterogeneity in Change
with Reasonable Tools

References

References

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